matplotlib Cheat sheet

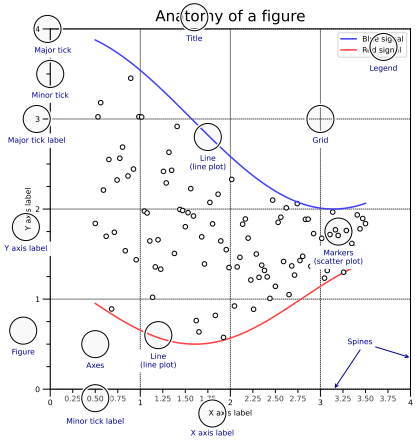
Version 3.10.8

Quick start

API

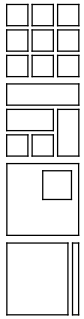
```
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt


X = np.linspace(0, 2*np.pi, 100)
Y = np.cos(X)

fig, ax = plt.subplots()
ax.plot(X, Y, color='green')

fig.savefig("figure.pdf")
plt.show()
```

Anatomy of a figure



Subplots layout

API



```
subplot[s](rows, cols, …)
fig, axs = plt.subplots(3, 3)

G = gridspec(rows,cols, …)
ax = G[0, :]

ax.inset_axes(extent)

d=make_axes_locatable(ax)
ax = d.new_horizontal('10%')
```

Getting help

- matplotlib.org
- discourse.matplotlib.org
- stackoverflow.com/questions/tagged/matplotlib

Basic plots

`plot([X], Y, [fmt], …)` API
X, **Y**, fmt, color, marker, linestyle

`scatter(X, Y, …)` API
X, **Y**, [s]izes, [c]olors, marker, cmap

`bar[h](x, height, …)` API
x, **height**, width, bottom, align, color

`imshow(Z, …)` API
Z, cmap, interpolation, extent, origin

`contour[f]([X], [Y], Z, …)` API
X, Y, **Z**, levels, colors, extent, origin

`pcolormesh([X], [Y], Z, …)` API
X, Y, **Z**, vmin, vmax, cmap

`quiver([X], [Y], U, V, …)` API
X, Y, **U**, **V**, C, units, angles

`pie(X, …)` API
Z, explode, labels, colors, radius

`text(x, y, text, …)` API
x, y, **text**, va, ha, size, weight, transform

`fill[_between][x](…)` API
X, Y1, Y2, color, where

Advanced plots

`step(X, Y, [fmt], …)` API
X, **Y**, fmt, color, marker, where

`boxplot(X, …)` API
X, notch, sym, bootstrap, widths

`errorbar(X,Y,xerr,yerr, …)` API
X, Y, xerr, yerr, fmt

`hist(X, bins, …)` API
X, bins, range, density, weights

`violinplot(D, …)` API
D, positions, widths, vert

`barbs([X], [Y], U, V, …)` API
X, Y, **U**, **V**, C, length, pivot, sizes

`eventplot(positions, …)` API
positions, orientation, lineoffsets

`hexbin(X, Y, C, …)` API
X, Y, C, gridsize, bins

Scales

API

`ax.set_[xy]scale(scale, …)`

linear any values

log values > 0

symlog any values

logit 0 < values < 1

Projections

API

`subplot(…, projection=p)`

p='polar'

p='3d' API

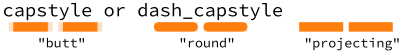
p=ccrs.Orthographic() API
import cartopy.crs as ccrs

Lines

API

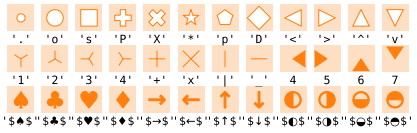
linestyle or ls

capstyle or dash_capstyle



Markers

API



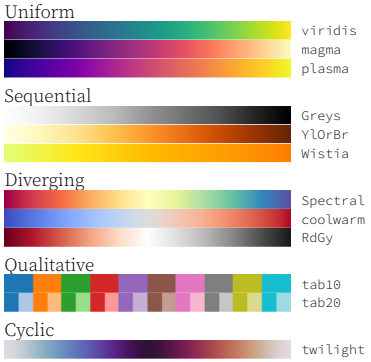
Colors

API

Colormaps

API

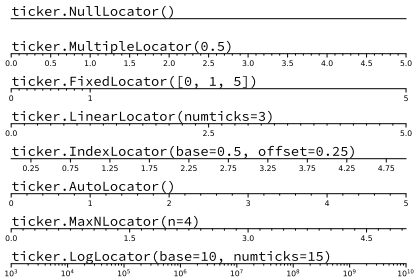
`plt.get_cmap(name)`



Tick locators

API

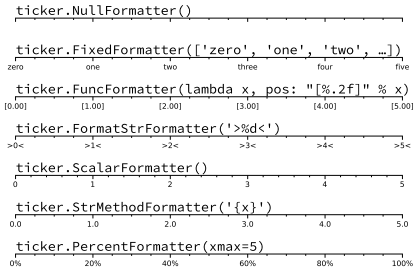
```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_locator(locator)
```



Tick formatters

API

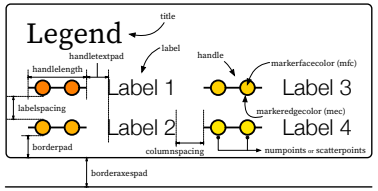
```
from matplotlib import ticker
ax.[xy]axis.set_[minor|major]_formatter(formatter)
```



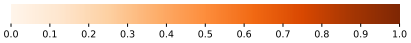
Ornaments

API

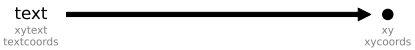
`ax.legend(…)`
handles, labels, loc, title, frameon



`ax.colorbar(…)` API
mappable, ax, cax, orientation



`ax.annotate(…)` API
text, **xy**, **xytext**, xycoords, textcoords, arrowprops



Event handling

API

```
fig, ax = plt.subplots()
def on_click(event):
  print(event)
fig.canvas.mpl_connect(
  'button_press_event', on_click)
```

Animation

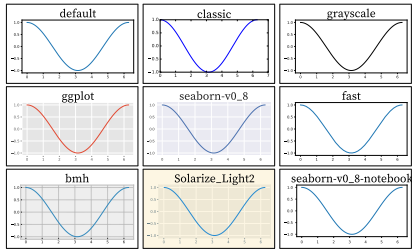
API

```
import matplotlib.animation as mpla

T = np.linspace(0, 2*np.pi, 100)
S = np.sin(T)
line, = plt.plot(T, S)
def animate(i):
    line.set_ydata(np.sin(T+i/50))
anim = mpla.FuncAnimation(
    plt.gcf(), animate, interval=5)
plt.show()
```

Styles

API

`plt.style.use(style)`



Quick reminder

```
ax.grid()
ax.set_[xy]lim(vmin, vmax)
ax.set_[xy]label(label)
ax.set_[xy]ticks(ticks, [labels])
ax.set_[xy]ticklabels(labels)
ax.set_title(title)
ax.tick_params(width=10, …)
ax.set_axis_[on|off]()

fig.suptitle(title)
fig.tight_layout()
plt.gcf(), plt.gca()
mpl.rc('axes', linewidth=1, …)
[fig|ax].patch.set_alpha(0)
text=r'$\frac{-e^{i\pi}}{2^n}$'
```

Keyboard shortcuts

API

ctrl + s	Save	ctrl + w	Close plot
r	Reset view	f	Fullscreen 0/1
f	View forward	b	View back
p	Pan view	o	Zoom to rect
x	X pan/zoom	y	Y pan/zoom
g	Minor grid 0/1	G	Major grid 0/1
l	X axis log/linear	L	Y axis log/linear

Ten simple rules

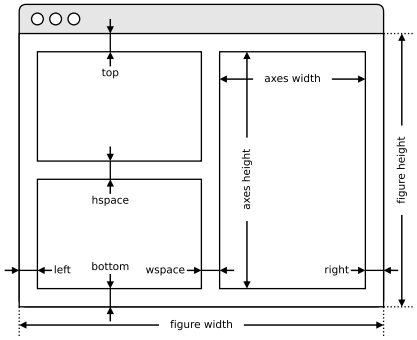
READ

1. Know your audience
2. Identify your message
3. Adapt the figure
4. Captions are not optional
5. Do not trust the defaults
6. Use color effectively
7. Do not mislead the reader
8. Avoid "chartjunk"
9. Message trumps beauty
10. Get the right tool

Axes adjustments

API

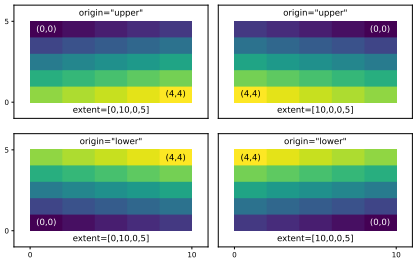
plt.**subplots_adjust**(...)



Extent & origin

API

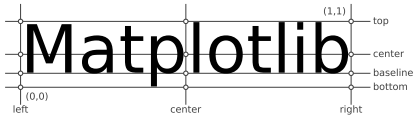
ax.**imshow**(extent=..., origin=...)



Text alignments

API

ax.**text**(..., ha=... , va=..., ...)



Text parameters

API

ax.**text**(..., family=..., size=..., weight=...)
ax.**text**(..., fontproperties=...)

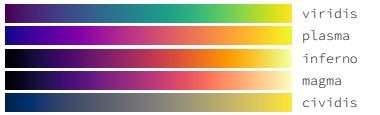
The quick brown fox — xx-large (1.73)
The quick brown fox — x-large (1.44)
The quick brown fox — large (1.20)
The quick brown fox — medium (1.00)
The quick brown fox — small (0.83)
The quick brown fox — x-small (0.69)
The quick brown fox — xx-small (0.58)

The quick brown fox jumps over the lazy dog — black (900)
The quick brown fox jumps over the lazy dog — bold (700)
The quick brown fox jumps over the lazy dog — semibold (600)
The quick brown fox jumps over the lazy dog — normal (400)
The quick brown fox jumps over the lazy dog — ultralight (100)

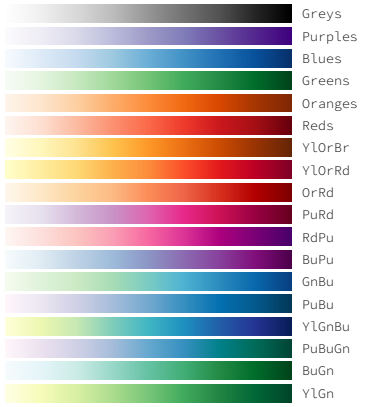
The quick brown fox jumps over the lazy dog — monospace
The quick brown fox jumps over the lazy dog — serif
The quick brown fox jumps over the lazy dog — sans
The quick brown fox jumps over the lazy dog — cursive

The quick brown fox jumps over the lazy dog — italic
The quick brown fox jumps over the lazy dog — normal

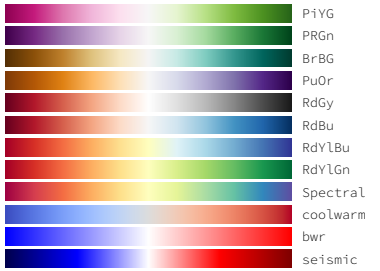
Uniform colormaps



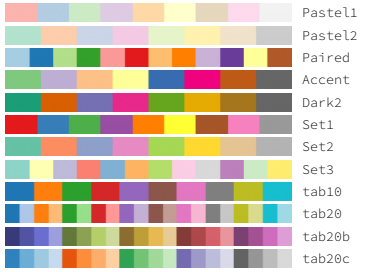
Sequential colormaps



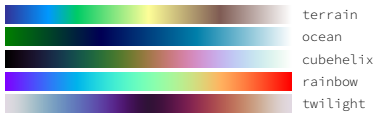
Diverging colormaps



Qualitative colormaps



Miscellaneous colormaps

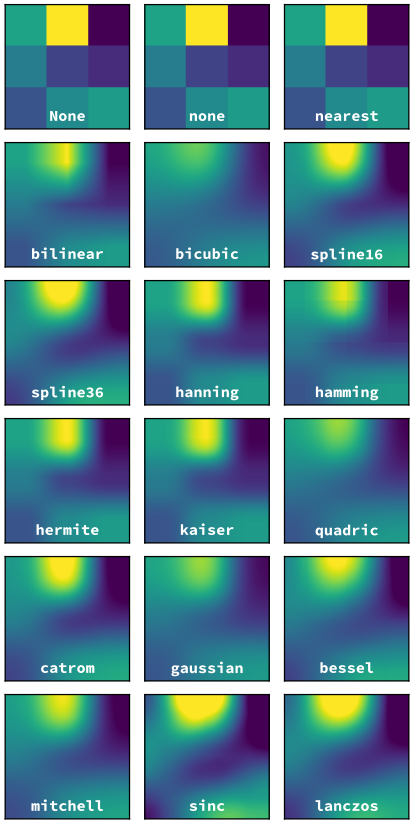


Color names

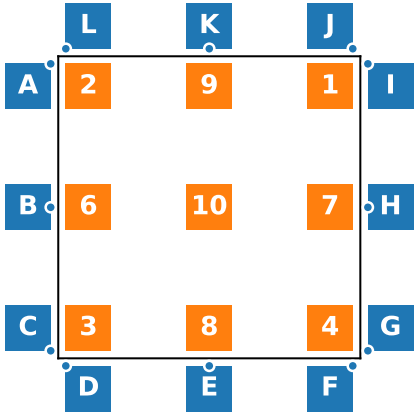
API

Image interpolation

API



Legend placement



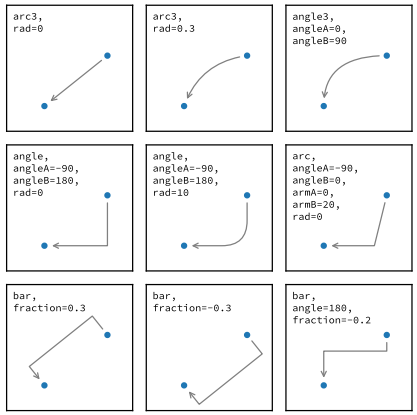
ax.**legend**(loc="string", bbox_to_anchor=(x, y))

2: upper left — 9: upper center — 1: upper right
6: center left — 10: center — 7: center right
3: lower left — 8: lower center — 4: lower right

A: upper right / `(-0.1, 0.9)`
B: center right / `(-0.1, 0.5)`
C: lower right / `(-0.1, 0.1)`
D: upper left / `(0.1, -0.1)`
E: upper center / `(0.5, -0.1)`
F: upper right / `(0.9, -0.1)`
G: lower left / `(1.1, 0.1)`
H: center left / `(1.1, 0.5)`
I: upper left / `(1.1, 0.9)`
J: lower right / `(0.9, 1.1)`
K: lower center / `(0.5, 1.1)`
L: lower left / `(0.1, 1.1)`

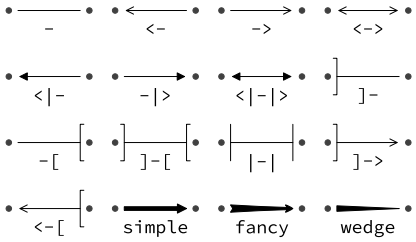
Annotation connection styles

API



Annotation arrow styles

API



How do I ...

... **resize a figure?**
→ fig.set_size_inches(w, h)
... **save a figure?**
→ fig.savefig("figure.pdf")
... **save a transparent figure?**
→ fig.savefig("figure.pdf", transparent=True)
... **clear a figure/an axes?**
→ fig.clear() → ax.clear()
... **close all figures?**
→ plt.close("all")
... **remove ticks?**
→ ax.set_[xy]ticks([])
... **remove tick labels ?**
→ ax.set_[xy]ticklabels([])
... **rotate tick labels ?**
→ ax.tick_params(axis="x", rotation=90)
... **hide top spine?**
→ ax.spines['top'].set_visible(False)
... **hide legend border?**
→ ax.legend(frameon=False)
... **show error as shaded region?**
→ ax.fill_between(X, Y+error, Y-error)
... **draw a rectangle?**
→ ax.add_patch(plt.Rectangle((0, 0), 1, 1)
... **draw a vertical line?**
→ ax.axvline(x=0.5)
... **draw outside frame?**
→ ax.plot(..., clip_on=False)
... **use transparency?**
→ ax.plot(..., alpha=0.25)
... **convert an RGB image into a gray image?**
→ gray = 0.2989*R + 0.5870*G + 0.1140*B
... **set figure background color?**
→ fig.patch.set_facecolor("grey")
... **get a reversed colormap?**
→ plt.get_cmap("viridis_r")
... **get a discrete colormap?**
→ plt.get_cmap("viridis", 10)
... **show a figure for one second?**
→ fig.show(block=False), time.sleep(1)

Performance tips

`scatter(X, Y)`	slow
`plot(X, Y, marker="o", ls="")`	fast
`for i in range(n): plot(i, X[i], "o")`	slow
`plot(X, marker="o", ls="")`	fast
`cla(); imshow(…); canvas.draw()`	slow
`im.set_data(…); canvas.draw()`	fast

Beyond Matplotlib

Seaborn: Statistical data visualization
Cartopy: Geospatial data processing
yt: Volumetric data visualization
mpld3: Bringing Matplotlib to the browser
Datashader: Large data processing pipeline
plotnine: A grammar of graphics for Python

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